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Site NDE Procedure - Ultrasonic Thickness Measurement, ATU-100

1.0 Scope and Application

This procedure establishes the requirements for ultrasonic thickness measurement of piping systems within the ATU-100. It applies to pre-startup baseline surveys and in-service NDE to monitor metal loss due to corrosion or erosion. The methodology ensures compliance with the inspection requirements for the Amine Treating Unit as mandated by the QSCP Hamilton Inspection Department.

All examination activities shall be performed in accordance with the technical requirements of ASME BPVC Section V. The scope includes all Class 1, Class 2, and Class 3 piping systems as defined by the site classification for sour gas, amine, and utility services. Douglas Ferreira is responsible for the technical oversight of these NDE activities.

2.0 Personnel Qualifications

Personnel performing ultrasonic thickness measurements shall be certified to at least Level II in the UT method in accordance with ASNT SNT-TC-1A. The examiner must demonstrate proficiency in the use of the ATU-100 instrument and be familiar with the specific calibration requirements for the materials encountered in the Hamilton Plant.

Douglas Ferreira, acting as the API 570 inspector, shall verify that all NDE technicians hold valid certifications and possess the necessary skills for accurate data acquisition. Gordon Lisle shall ensure that the engineering requirements for the unit are communicated to the inspection team prior to the commencement of site work.

3.0 References and Standards

The inspection intervals and data analysis methods are governed by API 570. Guidance for the identification of corrosion circuits and the selection of thickness measurement locations is provided by API 574. These standards form the basis for the asset integrity program at the Hamilton Plant.

Design and construction considerations for the piping systems are referenced from ASME B31.3. All NDE procedures and calibration blocks used during the examination shall meet the minimum requirements specified in ASME BPVC Section V to ensure repeatability and accuracy of the thickness readings.

4.0 Data Analysis and Retirement Criteria

The retirement thickness for any piping component is defined as the t-min specified in the line facts. Any reading obtained below this t-min shall be immediately raised as an NCR. The corrosion rate is determined by selecting the larger of the long-term rate, calculated since the first reading, and the short-term rate, calculated over the latest pair of readings at least 3 years apart.

Remaining life is calculated as (t - t-min) / rate. The next thickness reading must be scheduled at no more than half the remaining life, subject to the class maxima. Gordon Lisle is responsible for reviewing these calculations to ensure the continued integrity of the ATU-100 pressure boundaries.

5.0 Inspection Intervals

Inspection frequencies are determined by the fluid service and class. Class 1 systems, including sour gas and regenerator overhead, require thickness and external visual inspections every 5 years. Class 2 systems, covering amine, reflux, and treated gas, require thickness inspections every 10 years and external visual inspections every 5 years.

Class 3 systems, such as steam and cooling water, are subject to both thickness and external visual inspections every 10 years. These intervals are the maximum allowable periods and may be reduced based on the remaining life calculations performed by Douglas Ferreira. All data must be documented and archived within the QSCP Hamilton Inspection Department records.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	1995-11-20	_____
Approved	Gordon Lisle	Unit Engineer ATU-100, P.Eng.	1995-11-20	_____